AWS Infrastructure Architecture

Confidential Enterprise Infrastructure Project · Terraform-managed environment, with planned extensions marked separately

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CLIENTS

End users
browsers & apps (HTTPS)

Amazon Route 53
DNS alias records
(managed outside this stack)

HTTPS

ENGINEERS

IAM Identity Center
SSO + MFA, per-account access (no IAM users)

CI/CD

Bitbucket Pipelines
OIDC federation to scoped IAM roles

AWS Cloud · workload account (one account per environment)

GLOBAL EDGE
CloudFront · ACM certs in us-east-1

PLANNED
AWS WAF
managed + rate rules
(staging / production)

CloudFront · API
caching disabled
TLS 1.2+ · HTTP/2 & 3

CloudFront · Admin web
static assets cached
HTTP → HTTPS redirect

AWS Certificate Manager
DNS-validated certs
(edge + regional ALB)

Static website + forms

CloudFront
CloudFront Function
URL rewrite

/api/*

S3 private bucket
Origin Access Control
versioned

Lambda function URL
IAM auth via OAC

SNS (encrypted)
→ e-mail notification

AWS Region · configurable via Terraform variable

Internet gateway

VPC · 3-tier subnet design across 2 Availability Zones

S3 gateway endpoint

Availability Zone A

Public subnet

Amazon ECS on Fargate · ARM64
api · worker · scheduler · admin-web
Circuit-breaker rollback · ECS Exec
Secrets injected at start · KMS-encrypted logs

Private-app subnet

CloudFront VPC origin · HTTPS

Internal ALB
HTTPS 443 only · TLS 1.3
SNI · host-based routing
target health checks

PostgreSQL
TLS · 5432

Private-data subnet (no internet route)

RDS PostgreSQL 18
gp3 · KMS-encrypted · not public
TLS enforced (rds.force_ssl)
IAM DB auth enabled
RDS-managed master secret

Automated backups · Multi-AZ, retention and deletion protection set per environment

Availability Zone B

Public subnet

PLANNED
NAT gateways
count is a variable:
0 in dev · 1–2 per env
(one per AZ in prod)
Dev runs tasks with public IPs (inbound from ALB only) to avoid NAT cost

Private-app subnet

No public IP — reachable only from CloudFront VPC origin SG

Private-data subnet (no internet route)

PLANNED
Multi-AZ standby
(production design)
+ cross-Region backups

REGIONAL SERVICES (Terraform-managed)

Amazon ECR
immutable tags · scan on push
KMS · lifecycle policy

Secrets Manager
write-only values (never in Terraform state)

AWS KMS
customer-managed key per env
automatic rotation

IAM
least-privilege task roles
OIDC roles for CI/CD

Amazon CloudWatch
logs · metric filters · alarms

Amazon SNS
alarm notifications

EventBridge Scheduler
office-hours start/stop
(ECS + RDS)

Amazon S3
private uploads bucket
KMS · TLS-only policy

AWS Budgets
monthly cost alerts

LEGEND

Implemented — defined in Terraform; dev environment deployed

Planned — in the architecture design, not yet built

Purple: CloudFront → internal ALB (VPC origin) · Dashed grey: DNS resolution

SECURITY CONTROLS (IMPLEMENTED)

- Security-group chaining: CloudFront VPC origin → ALB → tasks → database
- Default security group locked (no rules) · admin web has no database path
- Encryption at rest with one rotating KMS key per environment
- Secrets injected by ECS; values written write-only, never in Terraform state
- IAM trust policies scoped with aws:SourceAccount and OIDC sub claims

PLANNED EXTENSIONS (DESIGN ONLY)

- Separate staging and production accounts from the same modules
- AWS WAF on CloudFront · NAT gateway per AZ · Multi-AZ RDS
- Cross-Region automated backups for disaster recovery
- Control Tower guardrails, GuardDuty, Security Hub, org CloudTrail